

EDA Report – Automobile Dataset

This Exploratory Data Analysis (EDA) report presents an overview of the Automobile dataset. The report details the cleaning steps, treatment of missing data, and visual analysis of vehicle prices, fuel efficiency, engine capacity, and manufacturer trends.

Data cleaning

The raw Automobile dataset contained redundant and inconsistent data.

Key cleaning steps included:

Removed columns: Dropped ``normalized-losses`` and ``symboling`` as they were not needed. Removed duplicate rows using ``drop_duplicates()``.

Missing values: Replaced with ``NaN`` and removed incomplete rows with ``dropna()``.

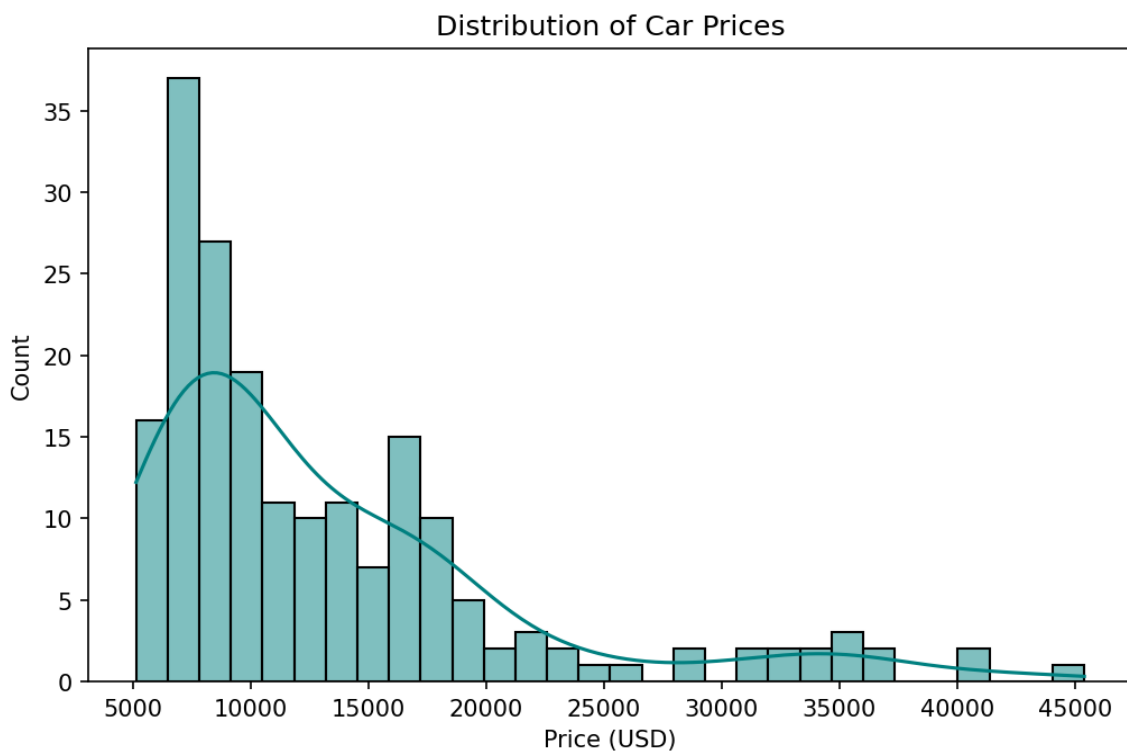
Data types: Converted numeric columns to ``int64`` using ``pd.to_numeric()``.

Column names: Standardised names to lowercase for consistency.

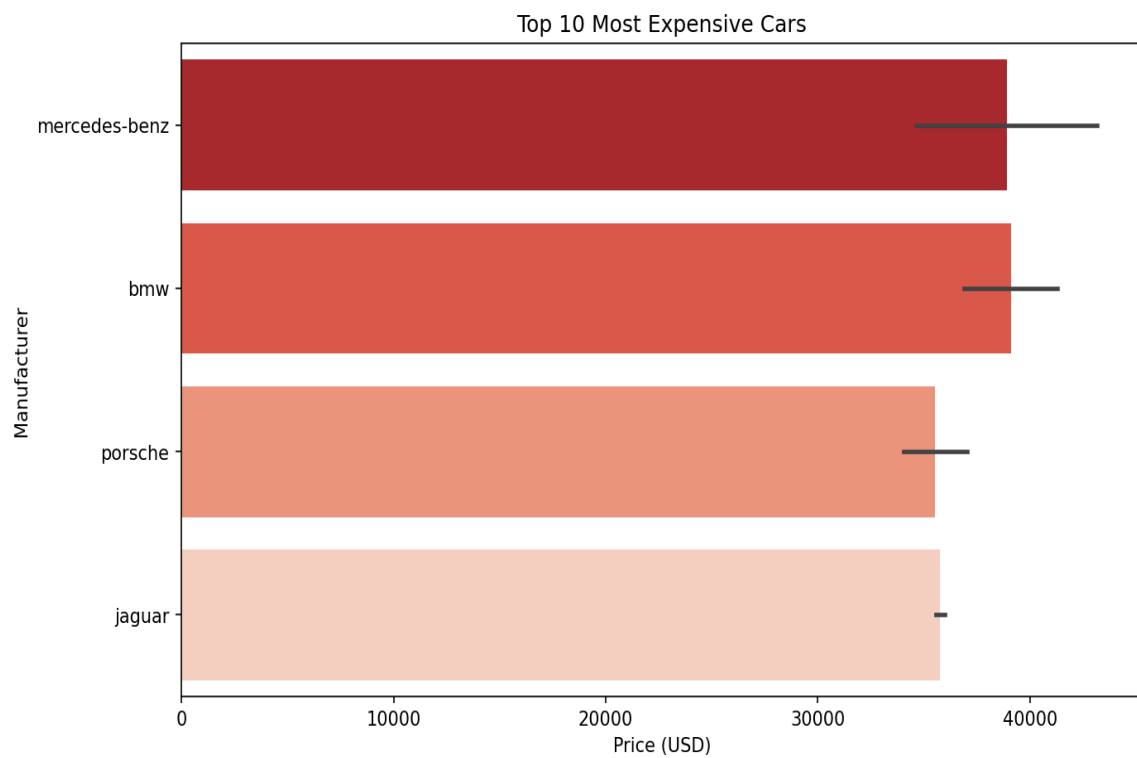
The cleaned dataset was then used for analysis and visualisations.

Missing Data

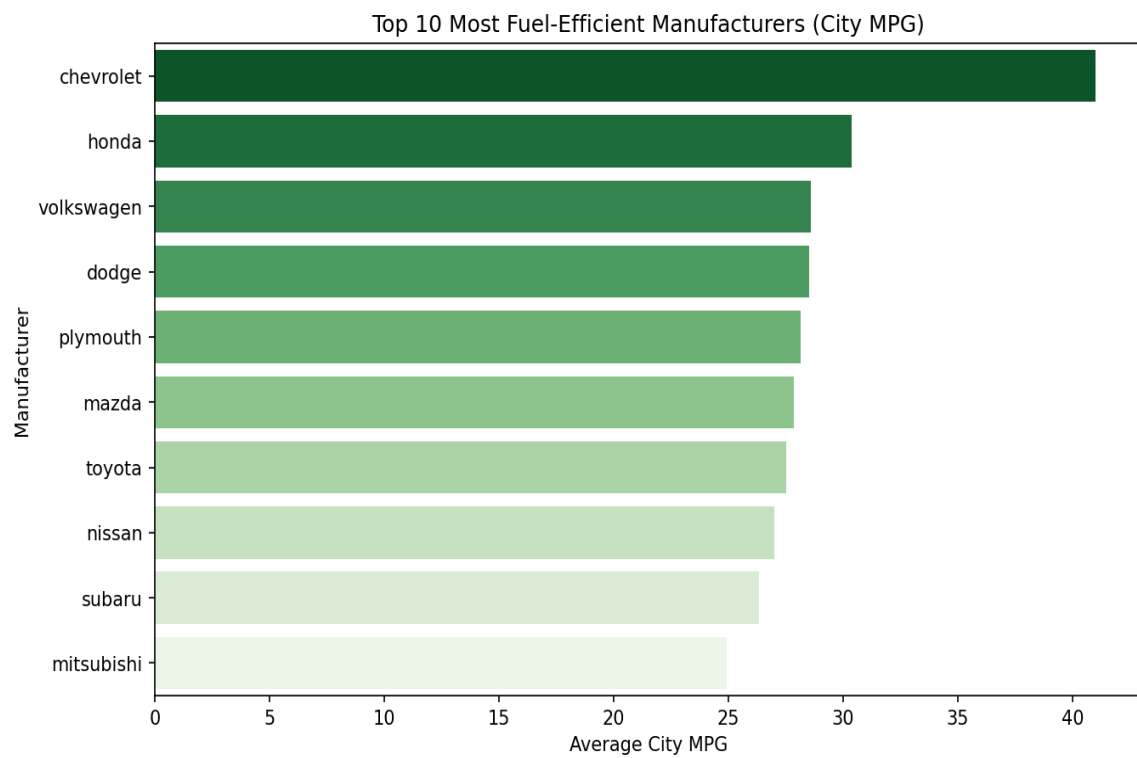
Some rows had missing entries for price, horsepower, and other numeric fields. To maintain data accuracy, all rows with missing values were removed. After cleaning, the dataset contained no missing values, ensuring reliable analysis.



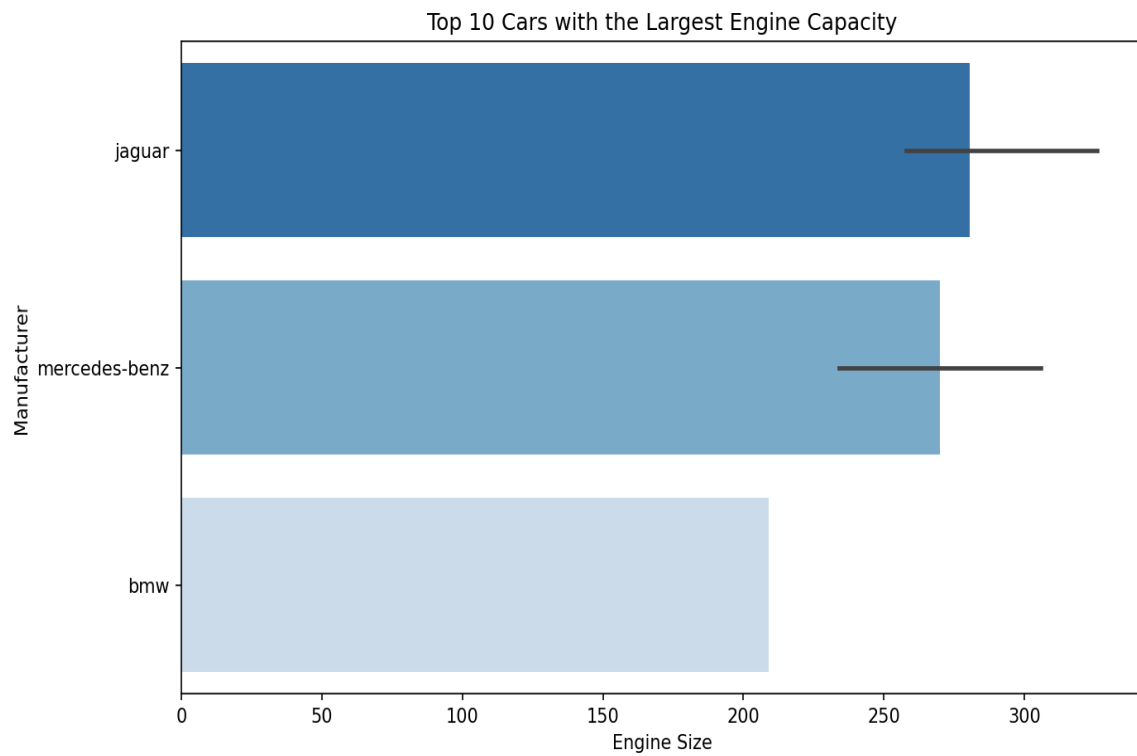
Distribution of Car Prices



Top 5 Most Expensive Cars

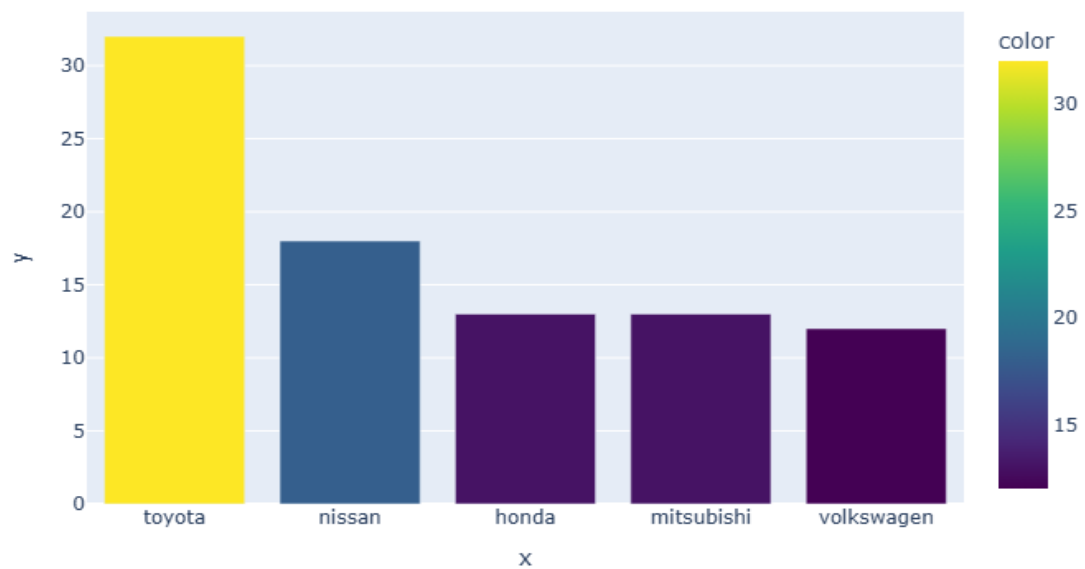


Top 10 Most Fuel-Efficient Manufacturers

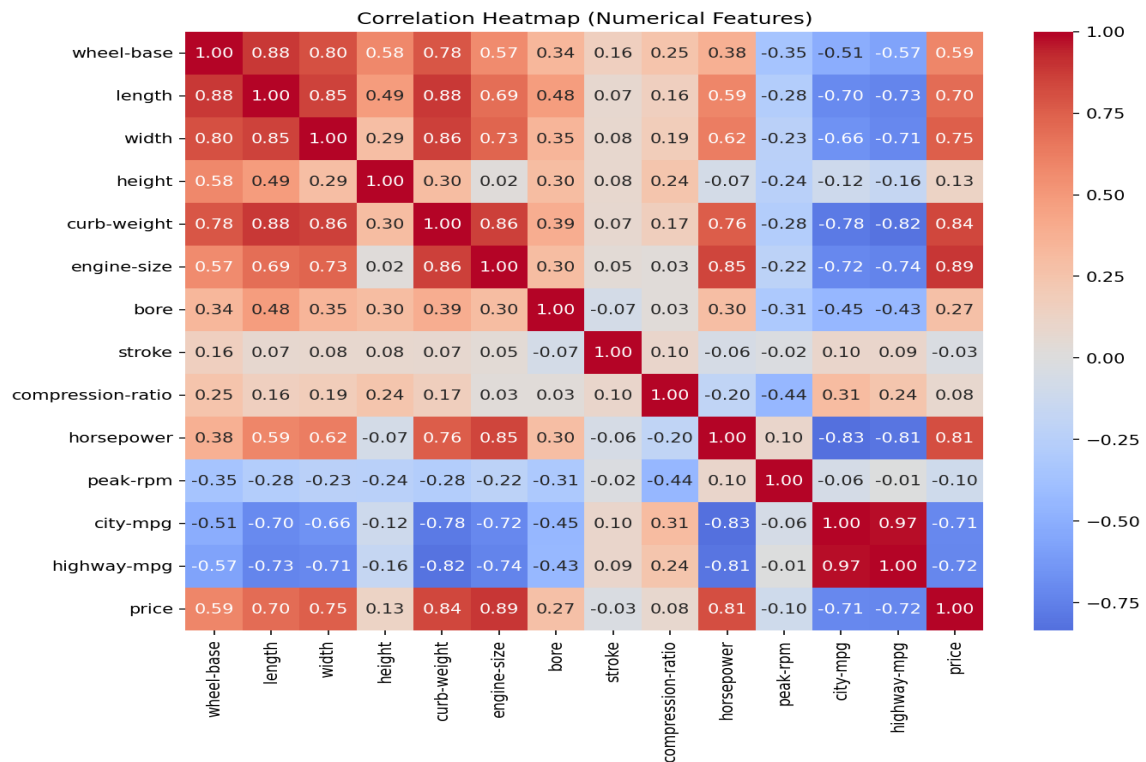


Top 10 Cars with Largest Engine Capacity

Top 5 Manufacturers by Number of Models



Top 10 Manufacturers by Number of Models



Correlation Heatmap of Numerical Features

In summary, this analysis indicates that engine size, horsepower, and curb weight are strong predictors of price. Manufacturers producing high-performance vehicles also dominate the upper end of the price spectrum. Meanwhile, smaller engines and lighter cars tend to be more fuel-efficient, as shown in the city MPG comparisons.

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